Aaryan Sharif

aaryan.sharif24@imperial.ac.uk
linkedin.com/in/aaryansharif · github.com/aary1n · +44 7917 979050 · London, UK

EDUCATION

Imperial College London – MEng Materials Science & Engineering Sep 2024 – Jun 2028 (expected)
Current average: 74.6% – First Class · Focus: Applied Mathematics & Computational Modelling
Relevant mathematics: Probability & Statistics – **100%** component · PDEs · Tensor & Vector Calculus

RESEARCH EXPERIENCE

Contractor, Anthropic Sep 2026 – Present

- Evaluating frontier-model swarm behaviour: built harnesses to stress-test multi-agent edge cases; wrote 6+ reproducible failure reports, each with a minimal repro and suggested fix, that engineers acted on.

Research Intern (UROP), Imperial College London – Oxborrow Group Jun 2026 – Present
First-author manuscript in preparation

- Built a Python-driven COMSOL finite-element model of the dielectric resonator used in pentacene masers, validated against a published device (frequency within **1.3%**, unloaded Q within **1%**).
- Derived the maser's thermal detuning margin: how much pump heating the cavity can absorb before the maser falls below threshold, and how that margin depends on cavity Q and spin linewidth; tested Python codebase with every input traced to its primary source.
- Ran research through bespoke agentic tooling: MCP servers for Zotero semantic search and GitHub, parallel subagent loops, and Lean formalisation of quantum-mechanical proofs; repurposed the resonator model as an RL environment.

Universal Model for Atoms, Meta FAIR – Preview Benchmarking Jul 2025 – Sep 2025

- Benchmarked UMA checkpoints against MACE and NEP interatomic potentials in PyTorch/ASE – force/energy convergence and failure modes; automated the HPC evaluation across 50+ configurations, cutting per-structure setup from ∼15 min to < 1 min.

Research Intern (UROP), Imperial College London – Wenman Group Jul 2025 – Aug 2025

- Deployed the NEP/GPUMD machine-learning potential stack to model Zr/ZrO_2 interfaces in nuclear-fuel cladding; built and open-sourced `zrbuildr` for parameterised slab/interface generation and lattice-matched supercell search.

SELECTED PROJECTS

interpotential – Mechanistic Interpretability of ML Interatomic Potentials Oct 2026 – Present
Manuscript in preparation

- With Prof. Arash Mostofi (Imperial): probing the internal representations of the UMA, Orb and MACE machine-learning interatomic potentials with mechanistic-interpretability methods.

SiPhON – C++/Python Numerical Simulation Library Jan 2026 – Present

- Implemented a C++/pybind11 shift-invert Arnoldi eigensolver (Eigen/Spectra) with zero-copy NumPy bindings and Monte Carlo propagation of fabrication tolerances to eigenvalue uncertainty.

Grapple – Real-Time ML Under a Hard Latency Budget Nov 2025 – Present

- Wrote the C#/.NET host: Python inference workers share frames via memory-mapped arenas with a FlatBuffers schema; a lock-free 120 Hz cursor loop extrapolates between ∼15 Hz tracking updates.
- Built the low-latency cursor path of a real-time gesture-control runtime (MediaPipe Hands): PyTorch-trained MobileNetV3 served via ONNX Runtime against a 5 ms latency target – INT8 **P50 0.83 ms / P95 1.61 ms**, ∼50% lower latency than FP32 under the same benchmark protocol.
- Designed sensitivity-guided static INT8 quantisation (27/53 convs, per-channel): naive full-graph INT8 inflated cursor-tracking MSE by 56%; probing conv sensitivity on a split disjoint from calibration and eval cut the degradation to **+1.4%**.

ACHIEVEMENTS

- **QuantiHack** – sponsored by Jane Street, Optiver & Anthropic: ranked **1st/800** by cumulative PnL across five trading days after identifying execution throughput as the bottleneck and parallelising order placement; qualified for the London finals.
- **Optiver AI Chessathon** – ranked **1st–2nd** for most of the event; withdrew before the finals.
- **Full-Tuition Academic Scholarship**, Imperial Department of Materials – merit award.
- **Robert Warner Bursary** (sole cohort recipient) · **Imperial College UROP Bursary** (1 of 60 college-wide) · **Department of Materials UROP Funding** (awarded twice) · **upReach Tech500 Associate**.

TECHNICAL SKILLS

Languages: Python · C++20 · CUDA · C#/.NET · Bash
Libraries & Tools: NumPy/SciPy · Polars · pandas · PyTorch · Linux · Git · PBS/HPC